

Umer Hussain

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LinkedIn — GitHub

Professional Summary

Final Year Software Engineering student at Bahria University with hands-on experience in AI, full-stack development, cloud computing, and DevOps. Experienced in building Large Language Model (LLM) applications, AI Agents, Retrieval-Augmented Generation (RAG) systems, and cloud-native applications using Python, Django, React, Flutter, Microsoft Azure, Docker, Kubernetes, LangGraph, and Google ADK. Passionate about developing intelligent, scalable AI solutions with expertise in machine learning, prompt engineering, and enterprise AI systems.

Education

B.E Software Engineering, Bahria University Karachi	2023 – Present
- Final Year Student - CGPA: 3.78 / 4.00	
FSc Pre-Engineering, Adamjee Govt. Science College	2021 – 2023
- Grade: A - Percentage: 79%	
Matriculation, Chiniot Islamia Public School	2019 – 2021
- Grade: A+ - Percentage: 96% - Gold Medalist	

Work Experience

Internship Bootcamp (Remote) — EARTech Information Technology, Istanbul	Jan 2026 – Feb 2026
- Worked with Microsoft Azure services including Azure OpenAI, Virtual Machines, and cloud infrastructure to deploy AI applications. - Explored Large Language Models (LLMs), prompt engineering, and Azure OpenAI integrations. - Studied advanced machine learning concepts including Catastrophic Forgetting and Elastic Weight Consolidation (EWC). - Completed cybersecurity labs using Hack The Box and collaborated remotely on cloud-based AI projects.	
Teaching Assistant (Advanced Python Course) — NED University of Engineering & Technology	May 2025 – Aug 2025
- Assisted Prof. Dr. Kashif Asrar in delivering an Advanced Python course, mentoring students in Python programming and full-stack web development. - Taught Django fundamentals, project architecture, authentication, REST APIs, database integration, and deployment best practices. - Authored a comprehensive step-by-step guide for building a production-ready full-stack Django application from scratch. - Guided students in integrating AI capabilities using Large Language Models (LLMs), email services (SMTP), and relational databases into Django applications. - Provided hands-on support during labs, debugging sessions, and project development while helping students adopt industry-standard development workflows.	

Technical Skills

- **Languages:** Python, JavaScript, C#
- **Frameworks:** Django, React, Flutter, ASP.NET, MERN Stack
- **AI Technologies:** Large Language Models (LLMs), AI Agents, Prompt Engineering, Retrieval-Augmented Generation (RAG), Google ADK, LangChain, LangGraph, Ollama, Azure OpenAI
- **Machine Learning:** Scikit-learn, MLP, Random Forest, Naive Bayes, NLP, Model Evaluation
- **Cloud & DevOps:** Microsoft Azure, Docker, Kubernetes, Virtual Machines
- **Databases & APIs:** PostgreSQL, REST APIs, JSON
- **Tools:** Git, GitHub, VS Code, Visual Studio, Google Colab, Hack The Box

Projects

LabourGO

Developed a cloud-native freelance service platform using Django, Flutter, Docker, and Kubernetes.

- Built scalable REST APIs and mobile application features for customers and service providers.
- Containerized the application using Docker and orchestrated deployments with Kubernetes.
- Developed an AI Monitor Agent for intelligent monitoring and automated system insights.
- Implemented authentication, booking, messaging, notifications, and service management modules.

Flexbook

Developed a full-stack social media platform using React and Django integrated with Agentic AI.

- Built an AI chatbot using Google ADK to provide intelligent user interaction and assistance.
- Designed agentic workflows capable of performing contextual and autonomous tasks.
- Implemented messaging, posts, reels, friend system, and scalable REST APIs.
- Designed optimized backend architecture and relational database models.

RahnumAI

Built an AI-powered smart LMS and analytics system.

- Developed ML models using MLP, Random Forest, and Naive Bayes for plagiarism detection and grade prediction.
- Designed performance tracking and mental health insight features.
- Improved prediction accuracy using ensemble learning techniques.

EnterpriseRAG

Built an enterprise Retrieval-Augmented Generation (RAG) system for intelligent document question answering.

- Developed an AI agent using LangGraph, Ollama, and Google ADK to retrieve knowledge from enterprise documents.
- Implemented document ingestion, semantic retrieval, prompt engineering, and contextual response generation.
- Integrated local LLMs with vector-based knowledge retrieval for accurate and efficient responses.
- Designed modular agent workflows for enterprise automation and AI-powered assistance.

Leadership & Community Service

- Ambassador and Active Participant at Hopefelt Foundation.
- Participated in community service initiatives including medical camps, beach cleaning drives, and plantation campaigns.
- Contributed to volunteer activities and Blood donation focused on environmental awareness and social welfare with SunoSunaoZindagi.

Achievements

- Merit Scholarship Holder (Fall 2025) with 4.00 / 4.00 GPA.
- Technofest On-Campus Speed Programming Runner-Up 2026.
- Selected for National-Level Technofest Competition.
- Gold Medalist in Matriculation (96%).

Certifications

- NED Certified Full Stack Developer (3-Months, 1.0 CPD)
- NED Certified Advanced Python Developer (4-Months, 1.0 CPD)
- HackerRank Certified Software Engineer
- Microsoft Certified C# Fundamentals
- freeCodeCamp — Backend Development and APIs